

AI-Driven Smart Energy Digital Twin for Industrial Plants

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This is an academic, fully software-based simulation. All plant, machine, and sensor data shown in this application is SYNTHETIC / SIMULATED. No real hardware, IoT devices, or factory deployments are involved.

Plant Energy Summary

Timestamp	2024-01-01T00:00:00
Total Power (Kw)	0.0
Total Energy (Kwh)	0.0
Average Efficiency	0.9
Energy Cost	0.0
Co2 Emission (Kg)	0.0
Active Machines	10 / 10
Active Anomalies	0

Machine Performance

machine_id	machine_type	status	load_percentage	power_kw	efficiency	health_score
Motor_1	Motor	Running	50.0	0.0	0.9	100.0
Motor_2	Motor	Running	50.0	0.0	0.9	100.0
Conveyor_1	Conveyor	Running	50.0	0.0	0.9	100.0
Compressor_1	Compressor	Running	50.0	0.0	0.9	100.0
Pump_1	Pump	Running	50.0	0.0	0.9	100.0
HVAC_1	HVAC	Running	50.0	0.0	0.9	100.0
Boiler_1	Boiler	Running	50.0	0.0	0.9	100.0
Cooling_System_1	Cooling	Running	50.0	0.0	0.9	100.0
Production_Line_1	ProductionLine	Running	50.0	0.0	0.9	100.0
Lighting_1	Lighting	Running	50.0	0.0	0.9	100.0

Detected Anomalies

severity	anomaly_score	probable_reason	temperature	vibration
CRITICAL	0.09798683195253066	Abnormal temperature rise (possible overheating); Elevated vibration (possible bearing wear/mechanical fault); Pressure spike	57.6	4.5
CRITICAL	0.087690474642374	Pressure spike; Efficiency drop	55.13	1.2
CRITICAL	0.08475921956082688	Pressure spike; Efficiency drop	54.98	2.1
CRITICAL	0.07590632022235277	Elevated vibration (possible bearing wear/mechanical fault); Pressure spike	48.24	4.8
CRITICAL	0.07388949395667233	Efficiency drop	43.05	1.5
CRITICAL	0.07286662756303952	Elevated vibration (possible bearing wear/mechanical fault); Pressure spike	50.45	4.2
CRITICAL	0.07283339421101975	Abnormal temperature rise (possible overheating); Pressure spike	57.09	2.3
CRITICAL	0.0721839581179693	Pressure spike; Efficiency drop	52.77	1.8
CRITICAL	0.07190866639243954	Elevated vibration (possible bearing wear/mechanical fault); Pressure spike	53.58	5.1

severity	anomaly_score	probable_reason	temperature	vibration
CRITICAL	0.07123595322171894	Elevated vibration (possible bearing wear/mechanical fault); Pressure spike	45.74	5.2

Predictive Maintenance

asset_id	machine_type	predicted_condition	risk_score	recommendation	vibration	temperature	efficiency	operating_hours
Asset_1	Lighting	Warning	36.6	Schedule a routine inspection within the next maintenance window.	1.665	51.26	0.934	3349.99
Production_Line_1	ProductionLine	Healthy	0.0	No action required. Continue routine monitoring.	0.508	25.82	0.858	1924.51
Boiler_1	Boiler	Healthy	0.0	No action required. Continue routine monitoring.	0.358	21.52	0.812	4768.38
Compressor_1	Compressor	Healthy	0.0	No action required. Continue routine monitoring.	0.43	23.83	0.8291	4476.91
Conveyor_1	Conveyor	Healthy	0.0	No action required. Continue routine monitoring.	0.728	20.08	0.8613	5202.44
Cooling_System_1	Cooling	Healthy	0.0	No action required. Continue routine monitoring.	0.393	24.14	0.8409	4874.79
Plant_1	HVAC	Healthy	0.0	No action required. Continue routine monitoring.	0.593	21.13	0.7906	5716.55
Motor_1	Motor	Healthy	0.0	No action required. Continue routine monitoring.	0.412	22.89	0.8899	4814.05
Motor_2	Motor	Healthy	0.0	No action required. Continue routine monitoring.	0.738	24.86	0.8865	3306.95
Pump_1	Pump	Healthy	0.0	No action required. Continue routine monitoring.	0.621	24.12	0.8515	1771.3

Optimization Results

Current Energy (Kwh)	0.0
Optimized Energy (Kwh)	0.0
Energy Saved (Kwh)	0.0
Savings (%)	0.0